

1. Why are Information Systems (IS) essential in organizations?

তথ্য ব্যবস্থা (Information Systems) কেন একটি প্রতিষ্ঠানে অপরিহার্য?

1. **Improved Decision Making** – IS provides accurate and timely data, which helps managers make better decisions.
2. **Increased Efficiency** – Automates repetitive tasks, reduces errors, and speeds up processes.
3. **Better Communication** – Enhances internal and external communication through email, instant messaging, and collaborative platforms.
4. **Data Management** – Stores, organizes, and recover large volumes of data easily and securely.
5. **Competitive Advantage** – Organizations that use IS strategically can gain an edge over competitors.
6. **Cost Reduction** – Reduces labor costs and operational expenses through automation.
7. **Customer Service** – Improves customer satisfaction through quick responses and personalized services.
8. **Innovation Support** – Encourages new product development, better marketing strategies, and improved services.

IS helps in storing data, managing business operations, and making smart decisions.

2. Why do systems analysts need to know who the stakeholders are?

1. To understand who will use the system
2. To know their needs and problems
3. To collect correct requirements
4. To avoid conflict and confusion
5. To make sure the system helps everyone

3. Who are the typical stakeholders in an information system and their roles?

1. Owners - fund the project
2. Users - anyone who interacts with the system
3. Designer - translate business needs into technical solutions
4. Builders - develops and builds the system
5. Analysts – defines/gather the business needs and requirements

Users (use the system), managers (guide use), owners (fund it), analysts (design it), developers (build it), customers (benefit from it), support staff (maintain it).

4. Please explain what the consequences are if an information system lacks a system owner.

1. **No clear direction** – Nobody sets goals or vision for the system.
2. **Lack of funding** – No one ensures money is available.
3. **No accountability** – Problems don't get solved quickly.
4. **Poor communication** – Developers and users get confused.
5. **System may fail** – The system may be ignored or shut down.

Without an owner, the system lacks direction, funding, and accountability – leading to confusion or failure.

5. What are the difference between internal users and external users? Give example

✓ Internal Users

People who work **inside the company** and use the information system every day.

Examples:

- Manager
- Accountant
- Employees

✓ External Users

People who are **outside the company** but still use the system.

Examples:

- Customers
- Delivery companies
- Suppliers

Difference	Internal Users	External Users
Location	Inside the company	Outside the company
Usage frequency	Use the system daily	Use it when needed
Control	Follow company rules	Not under company rules
Examples	Manager, Staff	Customer, Supplier

6. What are the differences between the role of system analysts and the role of the rest of the stakeholders? ★★★

System Analyst:

A system analyst studies the problems in the current system, talks to users, finds the requirements, and designs a better system.

Key Roles:

- Understands the problem.
- Talks with users and gathers needs.
- Designs solutions (new systems).
- Works with developers to build the system.

Other Stakeholders:

And the rest of the stakeholders (e.g. owners, users, managers, customers) use the system, give feedback, pay, or help make important decisions..

Types and Roles:

- **System Owner** – Approves budgets and goals.
- **Users** – Use the system, give feedback.
- **Managers** – Make decisions based on system results.
- **Customers/Suppliers** – Use the system from outside (if needed).

Simple Table:

Point of Comparison	System Analyst	Other Stakeholders
Main Role	Studies and designs the system	Use, fund, or are affected by the system
Responsibility	Understand needs, create solutions	Provide requirements, feedback, and approvals
Technical Knowledge	Must know IT and system design	Not always technical; more focused on business needs
Involvement Level	Deeply involved throughout the project	Varies – some give input, others only approve or use

7.What kind of knowledge and skills should a system analyst possess?

Skills a System Analyst Needs:

1. Technical Skills:

- **System Design:** Understanding system setup.
- **Programming:** Knowledge of coding languages (Java, Python).
- **Database Management:** Managing databases (SQL).
- **Networking:** Basic network knowledge.
- **Software Tools:** Familiar with tools like UML and ERD.

2. Business Skills:

- **Problem Solving:** Finding technical solutions to business problems.
- **Requirements Gathering:** Collecting stakeholder needs.
- **Project Management:** Managing projects and deadlines.

3. Interpersonal Skills:

- **Communication:** Clear interaction with both tech and non-tech people.
- **Teamwork:** Collaborating with team members.
- **Negotiation:** Handling changes and solutions with stakeholders.

8.In addition to the business an computing knowledge that **system analysis** should possess, what are the **other essential skills** that they need to effectively complete their jobs?

1. Critical thinking ability.
2. Strong problem-solving capacity.
3. High-level written and verbal communication skills.
4. Project management skills.
5. Ability to work under pressure and to tight deadlines.
6. Knowledge of data modelling and data visualization tools.

9.Why are good interpersonal communication skills essential for system analysis?

- **Understanding needs:** Gather clear requirements.
- **Teamwork:** Makes it easier to work with others.
- **Getting feedback:** Helps in listening to feedback and improving the system.
- **Solving problems:** Makes it easier to find solutions.

- **Clear explanations:** Helps explain the system to everyone, even if they are not technical.

Main Theme in Bengali:

সিস্টেম অ্যানালিস্টদের ভালো যোগাযোগ দক্ষতা

কাজ বোঝা,

দলের সঙ্গে কাজ করা এবং

সমস্যা সমাধান করতে সাহায্য করে।

10.What are some of the business drivers for today's information systems?

Through the use of information systems

1. **Efficiency:** Saves time and costs.
2. **Competitive Advantage:** Helps stay ahead in the market.
3. **Customer Satisfaction:** Improves service.
4. **Data-Driven Decisions:** Helps in better decision-making.
5. **Globalization:** Supports global operations.
6. **Innovation:** Enables new products and services.
7. **Compliance:** Ensures legal requirements are met.

1. দক্ষতা (Efficiency): তথ্য সিস্টেম ব্যবহারের মাধ্যমে সময় এবং খরচ বাঁচানো যায়।
2. প্রতিযোগিতামূলক সুবিধা (Competitive Advantage): সিস্টেম ব্যবহারে ব্যবসা প্রতিযোগিতায় এগিয়ে থাকতে পারে।
3. গ্রাহক সন্তুষ্টি (Customer Satisfaction): তথ্য সিস্টেম গ্রাহক সেবা উন্নত করতে সহায়তা করে।
4. তথ্যভিত্তিক সিদ্ধান্ত গ্রহণ (Data-Driven Decisions): সঠিক তথ্য ব্যবহার করে ব্যবসা ভালো সিদ্ধান্ত নিতে পারে।
5. বিশ্বব্যাপী কার্যক্রম (Globalization): তথ্য সিস্টেম ব্যবসাকে আন্তর্জাতিক পর্যায়ে পরিচালনা করতে সহায়তা করে।
6. উদ্ভাবন (Innovation): নতুন পণ্য বা সেবা তৈরি করতে সিস্টেম সাহায্য করে।
7. আইনগত মান (Compliance): সিস্টেম ব্যবহারে ব্যবসা আইনি নিয়ম-নীতি অনুসরণ করতে পারে।

11.Difference Between E-Commerce and E-Business (with Bengali meaning)

Point	E-Commerce	E-Business
Meaning	Buying/selling online	All online business activities
Focus	Only on transactions	Full business process
Scope	Part of e-business	Bigger than e-commerce
Example	Buying a book on Amazon	Amazon managing stock, delivery, support

13.What are the most important technology drivers for today's information systems? (ABCC IM)

1. **Internet and Web** – Connects people and systems easily.
2. **Mobile Technology** – Access from anywhere using phones/tablets.
3. **Cloud Computing** – Stores and runs systems online.
4. **Big Data and Analytics** – Helps analyze large data for smart decisions.
5. **Artificial Intelligence (AI)** – Automates tasks and improves performance.
6. **Cybersecurity** – Protects systems and data.

14. What are the four steps in a system development process? What happens in each step?

✓ 1. Planning

- Set goals for the new system.
- Decide what problem to solve and whether it's worth solving.

✓ 2. Analysis

- Study the current system.
- Collect user needs and system requirements.

✓ 3. Design

- Create blueprints for the system (like diagrams, layout, data flow).
- Decide how the system will look and work.

✓ 4. Implementation

- Build and test the system.
- Install the system and train users.

15. Why is system initiation essential in the system development process?

System initiation is essential because it is the **first step** that helps to:

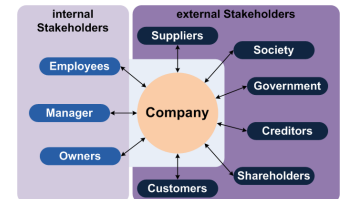
- Identify the **problem** or opportunity.
- Decide whether a **new system** is really needed.
- Set the **goals** and **scope** of the project.
- Get **approval** and **budget** to start the system development.

✓ **Main Point:** Without system initiation, the project may go in the wrong direction or fail to meet real needs.

****1. Assume you are a system analyst who will be conducting a requirements analysis for an individually owned brick-and-mortar retail store with a point-of-sale system. Identify who the typical internal and external users might include.

✓ Internal Users:

1. **Store Owner** – Manages the business, sets goals, checks sales reports.
2. **Cashiers** – Use the POS system for billing and transactions.
3. **Inventory Staff** – Track and manage stock levels.
4. **Store Manager** – Oversees operations, staff, and reports.



✓ External Users:

1. **Customers** – Buy products, receive receipts, may use loyalty points.
2. **Suppliers/Vendors** – Provide goods and check stock requirements.
3. **Technical Support** – External team who maintains or fixes the POS system.
4. **Accountants (if outsourced)** – Use sales data for financial tasks.

2. Assume you are a systems analyst for a consulting company and have been asked to assist the chief executive officer (CEO) of a regional bank. The bank recently implemented a plan to reduce the number of staff, including loan officers, as a strategy to maintain profitability. Subsequently, the bank has experienced chronic problems with backlogged loan officers who are able to review and approve or disapprove loans. The CEO of the bank is interested in solutions that would allow the approval process to move faster without increasing the number of loan officers, and has engaged your company to come up with suggestions. What is one type of system that you might recommend to the bank?

Recommended System:

Loan Management System (LMS) with automation and decision support.

Main Features:

1. **Automated Application Check:**
 - The system checks documents and basic eligibility automatically.
2. **Credit Score Analysis:**
 - It connects with credit databases to quickly review the applicant's score.
3. **Workflow Automation:**
 - Automatically sends loan requests to the right officer or process step.
4. **Document Management:**
 - Allows digital upload of files to reduce paperwork.

স্বয়ংক্রিয় আবেদন যাচাই:

- ◆ কাজ: কাগজপত্র ও যোগ্যতা অটো চেক করে।
- ◆ উপকার: সময় বাঁচে, ভুল কম হয়।

ক্রেডিট স্কোর বিশ্লেষণ:

- ◆ কাজ: ক্রেডিট ডেটাবেস থেকে স্কোর যাচাই করে।
- ◆ উপকার: ঝুঁকি কমে, ভালো গ্রাহক শনাক্ত হয়।

ওয়ার্কফ্লো অটোমেশন:

- ◆ কাজ: আবেদন সঠিক অফিসারে পাঠায়।
- ◆ উপকার: প্রক্রিয়া দ্রুত হয়, দেরি কমে।

ডকুমেন্ট ম্যানেজমেন্ট:

- ◆ কাজ: কাগজপত্র ডিজিটালি সংরক্ষণ করে।
- ◆ উপকার: কাগজ কম লাগে, হারানোর ভয় নেই।

নোটিফিকেশন ও ট্র্যাকিং:

- ◆ কাজ: আবেদনকারীদের লোন অবস্থা জানায়।
- ◆ উপকার: গ্রাহক সন্তুষ্ট বাড়ে, দেরি কম হয়।

5. Notifications & Tracking:

- Tracks loan status and sends updates to avoid delays.

3. How do communication and collaboration systems efficiency and effectiveness? What are some of the communication and collaboration systems that are being used by an increasing number of organizations?

How they improve efficiency and effectiveness:

- **Fast Communication** – Quick exchange of information (e.g., chat, video calls).
- **Easy Teamwork** – Teams can work together even from different locations.
- **Saves Time** – Less need for meetings and emails; tasks move faster.
- **File Sharing** – Easily share documents and files with others.

Popular Communication & Collaboration Systems:

- **Microsoft Teams** – Chat, video call, and file sharing in one place.
- **Slack** – A messaging app for quick team communication.
- **Zoom** – Used for video meetings and webinars.
- **Google Meet** – Video calls directly from your browser.
- **Trello / Asana** – Task and project management tools.
- **Google Drive / OneDrive** – Store and share documents online.

4. Identify the type of information system that clerical workers in an organization would typically use and why?

Clerical workers typically use a **Transaction Processing System (TPS)**.

✓ **Reason:**

- Helps with data entry, payroll, and billing.
- Handles many transactions quickly.
- Used for everyday office work.
- Simple for non-technical staff.
- **Examples:** Excel, billing software, payroll systems.

5. ethical issues in information systems:

1. **Privacy** – Using someone's personal data without asking.
2. **Stealing Data** – Taking secret information without permission.
3. **Wrong Use** – Using data to harm or cheat people.
4. **Hacking** – Breaking into computers or systems.
5. **False Data** – Putting in or spreading wrong information.
6. **Unfair Systems** – Systems that treat people unequally.

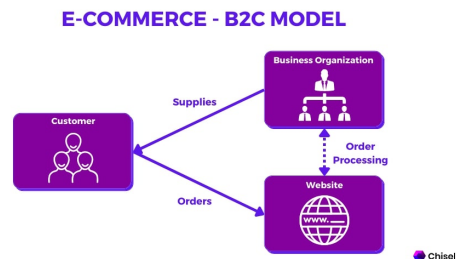
6. What are business to consumer (B2C) and business to business (B2B) web applications and what are some examples of each type?

Business to Consumer (B2C):

These are websites where **businesses sell products or services directly to customers** (individual people).

Examples:

- Amazon – for buying goods online
- Netflix – for watching movies/shows
- Daraz – for shopping

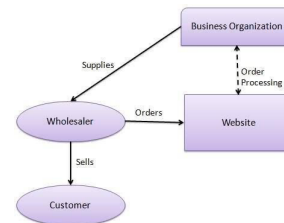


Business to Business (B2B):

These are websites where **businesses sell products or services to other businesses**.

Examples:

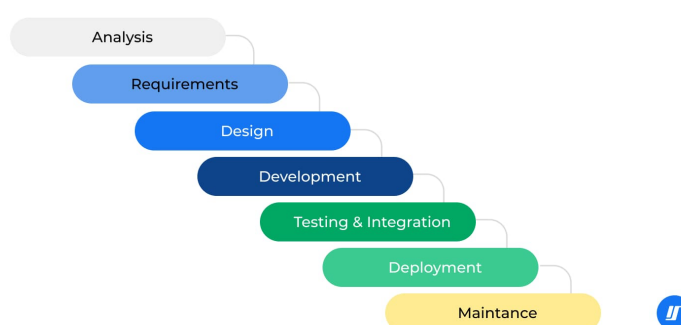
- Alibaba – for wholesale buying
- Salesforce – for business software (CRM)
- Microsoft Azure – cloud services for companies



7. While system development processes and methodologies can vary greatly, identify and briefly explain the generic phases of the system development process that are described in the textbook and which must be completed for any project. You are a contractor with a systems integration company.

✓ Main Steps in System Development (Easy Version):

1. **Planning** – Decide what the system should do and set goals.
2. **Analysis** – Find out what users need.
3. **Design** – Plan how the system will look and work.
4. **Implementation** – Build and test the system.
5. **Maintenance** – Fix problems and update the system after it's running.



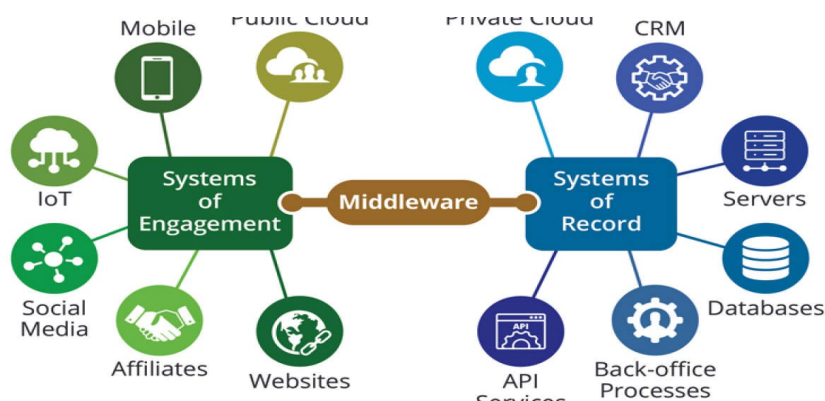
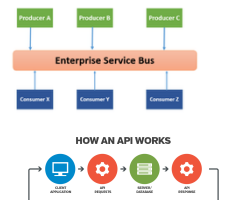
8. Your company has a contract with a local firm to link all of their systems so they can transparently work together. Their applications include a number of existing legacy systems, which were built at different times by different developers using a variety of languages and platforms, as well as several newer contemporary applications. What is the term for this type of linking? What type of tool would you most likely use and what are some examples of these tools?

✓ **Answer:**

- **Term: System Integration**
This is the process of linking different systems, both old (legacy) and new, to work together seamlessly.
- **Tool Used: Middleware**
Middleware helps different systems communicate and share data, especially when they use different technologies.

Examples of Middleware Tools:

- **Enterprise Service Bus (ESB)** – যেমন Apache Camel, MuleSoft
Facilitates data exchange between different systems
- **Application Programming Interfaces (APIs)** – যেমন REST API, SOAP
Allows one application to talk to another.
- **Integration Platforms** – যেমন Dell Boomi, Zapier, Microsoft Power Automate
Automatically sends data or performs tasks from one system to another.



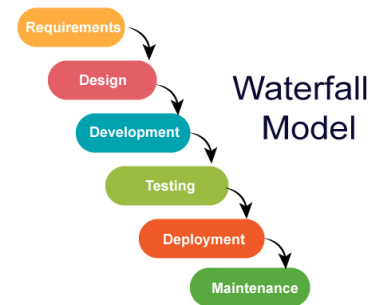
9. Your company has asked you to develop a new Web-based system to replace its existing legacy system. There will be very **little change** in business requirements and functionality from the existing legacy system. Suggest which system development process you might use and why.

You can use the **Prototyping Model** or **Agile Methodology**, but since the business requirements are stable and similar to the legacy system, the best choice is:

✓ System Development Process: Waterfall Model

◆ Why Use Waterfall Model:

- Requirements are **clear and fixed**
- Business functions will **not change much**
- Easier to plan and manage in **phases**
- Suitable for **replacing old systems** step by step.



Agile methodology হলো একটি পদ্ধতি যেখানে সিস্টেমটি ছোট ছোট সাইকেলে তৈরি করা হয়।

প্রক্রিয়া:

1. **পরিকল্পনা:** সিস্টেমের পরবর্তী অংশের জন্য পরিকল্পনা করা হয়।
2. **ডিজাইন এবং ডেভেলপমেন্ট:** নির্দিষ্ট সময়ে কাজ শেষ করা হয়।
3. **ফিডব্যাক:** শেষ হলে গ্রাহক বা স্টেকহোল্ডারের কাছ থেকে প্রতিক্রিয়া নেওয়া হয়।
4. **পরিবর্তন:** ফিডব্যাক অনুযায়ী পরিবর্তন করা হয়।
5. **পুনরাবৃত্তি:** এই প্রক্রিয়া চলতে থাকে যতক্ষণ না সিস্টেম সম্পূর্ণ হয়।

মূল বৈশিষ্ট্য:

- ফ্লেক্সিবল এবং **ফিডব্যাক** অনুযায়ী পরিবর্তন করা যায়।
- **গ্রাহকের মতামত** গ্রহণ করা হয়।
- দ্রুত এবং **ছোট ছোট অংশে** কাজ সম্পন্ন করা হয়।

10. You recently Joined a retail sales company which has recently bought out and assimilated a commercial industrial supply house. You have been asked to lead a project to develop a consolidated inventory-tracking system. Suggest which system development process you might use and why.

For this project, the **Agile Development Model** would be best. Here's why:

1. **Flexibility:** Agile allows for changes during development, which is useful as the company integrates the two businesses.
2. **Frequent Feedback:** Regular updates from stakeholders help make sure the system meets everyone's needs.
3. **Quick Delivery:** Agile delivers the system in small, working parts, allowing for fast improvements and adjustments.
4. **Adaptable:** As requirements evolve, Agile can easily incorporate changes to the inventory system.

ছোট একটা পার্ট নিয়ে কাজ করা হয় এবং ঐটাই development করা হয় ,তারপর stakeholders দেব দেওয়া হয় তারা তাদের feedback জানায় then সে অনুযায়ী কাজ হয়...

11. Your company president sits down beside you just before a meeting is to begin and tells you that people keep saying the customer needs to install a CRM, but doesn't really know what it is. The company president then asks you to explain it in nontechnical terms in the next 30 seconds.

CRM হচ্ছে একটি সফটওয়্যার সিস্টেম যা কোম্পানিকে গ্রাহকদের সঙ্গে সম্পর্ক স্থাপন, পরিচালনা এবং শক্তিশালী করতে সাহায্য করে। এটি গ্রাহকদের তথ্য, কেনাকাটার ইতিহাস এবং অন্যান্য যোগাযোগ সংক্রান্ত ডেটা সংরক্ষণ করে, যাতে কোম্পানি গ্রাহকদের আরও ভালো সেবা দিতে পারে এবং তাদের সঙ্গে সম্পর্ক মজবুত করতে পারে। সহজভাবে বলতে গেলে, এটা একটি ডিজিটাল অড্রেস বুক যা কোম্পানির গ্রাহকদের জন্য সঠিক তথ্য প্রদান করে।

15. What are two most important advantages of object-oriented software technologies over structured software technologies.

- **Faster development:** Reuse enables faster development. Object-oriented programming languages come with rich libraries of objects, and code developed during projects is also reusable in future projects.
- **Lower cost of development:** The reuse of software also lowers the cost of development.